

GFDL Vortex Tracker Format Instructions for WP-MIP

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This document describes the data format generated by the GFDL Vortex Tracker for sharing with the [WP-MIP](#) community for the Subproject 1: Tropical Cyclones. The format is similar to that of [the Automated Tropical Cyclone Forecasting \(ATCF\) system](#), developed by the US Navy, but please note that the format is not identical to ATCF. The output files are saved in a comma-separated ASCII format. To facilitate ease of use, unused or blank columns from the original Tracker output have been removed. The document below details the format, including information on file names and the corresponding data fields.

Filenames:

GFDL_Tracker_`\${region}``\${storm\_ID}``\${year}``\${model\_info}``\${initial_time}`.txt

`\${region}`: a two character index to indicate the basin/region of a TC:

- al: Atlantic Basin
- wp: Western North Pacific basin
- cp: Central North Pacific basin
- ep: Eastern North Pacific basin
- io: North Indian Ocean
- sh: Southern Hemisphere

`\${storm\_ID}`: a two digit index to indicate the annual cyclone number of a TC

e.g. [2024 TC number and names in the Atlantic basin](#)

`\${year}`: a four digit index for the year of a TC; this should be “2024” for WP-MIP.

`\${model\_info}`: a four character index to indicate the model (first two characters), the model's category (third character: physical, AI, or hybrid), and the use of initial condition (fourth character: OIC or SIC), for example:

- shpo = GFDL, physical model, OIC
- shps = GFDL, physical model, SIC
- ecpo = ECMWF, physical model, OIC
- echo = ECMWF, hybrid model, OIC
- ecao = ECMWF, AI model, OIC

Please check the complete list in the introduction of the published dataset.

`${initial_time}`: a ten digit index to indicate the initial date and time as “yyyymmddhh”

Fields in the data:

There are 34 common fields. Non-defined values are set to “-999” or “-999.99” if there is no specific definition for each field.

Fields 1-20: BASIN, CY, DATE/TIME, MODEL, LEADTIME, LatN/S, LonE/W, VMAX, MSLP, RAD, WINDCODE, RAD1, RAD2, RAD3, RAD4, POUTER, ROUTER, RMW, DIR, SPEED

1. BASIN

AL: Atlantic Basin

WP: Western North Pacific basin

CP: Central North Pacific basin

EP: Eastern North Pacific basin

IO: North Indian Ocean

SH: Southern Hemisphere

2. CY: annual cyclone number

3. DATE/TIME: initial time of the forecast; format: YYYYMMDDHH

4. MODEL: same four character index as the `${model_info}` in the filename

5. LEADTIME: forecast lead time, from 000 to 240.

6. LatN/S: Latitude; 0 - 900 tenths of degrees, N/S is the hemispheric index.

7. LonE/W: Longitude; 0 - 1800 tenths of degrees, E/W is the hemispheric index.

8. VMAX: Maximum sustained wind speed in “m/s”.

9. MSLP: Minimum sea level pressure in “hPa”.

Fields 10-15: Metrics for the critical radii of 34-, 50- and 64-kt winds. For a TC weaker than 34 kts, Fields 12-15 will be listed as “-999.99”. For a TC with VMAX exceeding 50 kts, multiple records could be listed for a single lead time, each describing different radii for 50-kt and 64-kt winds. All other fields will be kept the same for each radius.

10. RAD: Wind intensity identifier in this record for the radii of 34, 50 or 64 kt.

11. WINDCODE: Radius code: NEQ indicates that the Fields 12-15 will hold values of wind radii in quadrants, with Field 12 holding the NE quadrant and continuing clockwise through Field 15.

12. RAD1: The radius of RAD in NE Quadrant. Unit: "km".
13. RAD2: The radius of RAD in SE Quadrant. Unit: "km".
14. RAD3: The radius of RAD in SW Quadrant. Unit: "km".
15. RAD4: The radius of RAD in NW Quadrant. Unit: "km".
16. POUTER: pressure in hPa of the last closed isobar.
17. ROUTER: radius of the last closed isobar in "km".
18. RMW: radius of max winds (Field 8: VMAX) in "km".
19. DIR: storm movement direction, 0 - 359 degrees.
20. SPEED: storm translation speed in "m/s".

Fields 21-26: Thermodynamic diagnostics of Cyclone Phase Space based on [Bob Hart's MWR paper](#), and also a check for a warm core in the 300-500 mb layer.

21. "THERMO PARAMS" marker identifies the beginning of the thermodynamic diagnostics portion of the record.
22. Cyclone Phase Space "Parameter B" for thermal asymmetry. (Values are *10)
23. Cyclone Phase Space lower level (600-900 hPa) thermal wind parameter, for diagnosing low-level warm core. (Values are *10)
24. Cyclone Phase Space upper level (300-600 mb) thermal wind parameter, for diagnosing upper-level warm core. (Values are *10)
25. "Y", "N" or "U" for warm core diagnostics in the 300-500 hPa layer. Value of "Y" is returned if a closed contour in the temperature field exists surrounding the 300-500 mb warm core, using a contour interval that is specified in Field 26. "N" is returned if a check was performed and a closed contour does not exist. "U" is returned if the check was unable to be performed, due to missing data, etc.
26. Value of temperature contour interval (in degK * 10) that is used for determining if a closed contour exists in the 300-500 hPa temperature field.

Fields 27-29: Diagnostics of 200-850 hPa vertical shear based on [the same method as used in SHIPS](#).

27. "SHR82" marker identifies the beginning portion of the record that reports parameters for the 200-850 hPa vertical shear.
28. Magnitude (m/s) of the 200-850 hPa vertical shear. The shear is calculated out to $r=500$ km from the center of the circulation as determined by the tracker at 850 hPa. Prior to computing the shear, the vortex is removed.
29. The direction (0-360 deg) of the 200-850 hPa shear. This is the direction that the shear is pointing to. This direction is used because that is also how SHIPS reports shear. So it's important to know that this is *NOT* the direction the shear is coming from.

Fields 30-31: Diagnostics of SST

30. "SST" marker identifies the beginning of the SST diagnostics portion of the record.
31. Gaussian-smoothed value of SST ($\text{degK} * 10$) surrounding the tracker-derived fix position for this lead time. The algorithm uses a Barnes analysis, with an e-folding radius of 125 km and a maximum radius of influence of 250 km. The value is set to "0" when the TC is over land, and set to "-999" if the surface temperature is not found in the input data.

Fields 32-34: Diagnostics of axisymmetric radius of maximum wind (ARMW). The computation for ARMW differs from that for the RMW in Field 18 above, where that is a "point value" of RMW indicating the distance from the tracker-derived fix position to the grid point where the max wind was found. For ARMW, the gridded data are interpolated into 3-km-wide radial bands surrounding the tracker-derived fix position, then azimuthally-averaged values are computed for each radial band, and then the distance from the fix position to the band with the highest mean wind value is returned as the ARMW, subject to constraints of gradients that are checked.

32. "ARMW" marker identifies the beginning of diagnostics for axisymmetric RMW.
33. Distance of the ARMW in "km".
34. The mean value of the wind speed in "m/s" in the band that was returned as the ARMW.